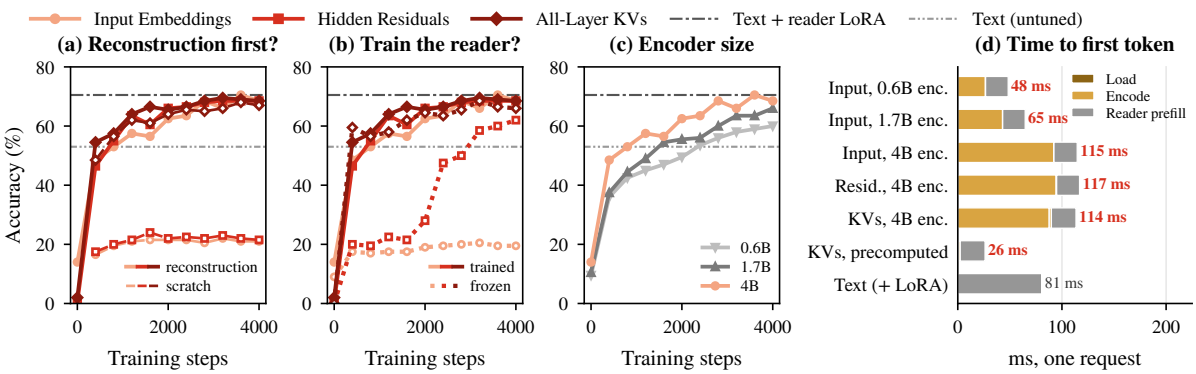
Input Embeddings
Hidden Residuals
All-Layer KVs
Text + reader LoRA
Text (untuned)
(a) Reconstruction first?
(b) Train the reader?
(c) Encoder size
(d) Time to first token
Accuracy (%)
Training steps
reconstruction
scratch
trained
frozen
0.6B
1.7B
4B
Input, 0.6B enc.
Input, 1.7B enc.
Input, 4B enc.
Resid., 4B enc.
KVs, 4B enc.
KVs, precomputed
Text (+ LoRA)
48 ms
65 ms
115 ms
117 ms
114 ms
26 ms
81 ms
Load
Encode
Reader prefill
ms, one request